

Expt. No. : 1

Date :

BUILD A PYTHON PROGRAM TO IMPLEMENT FIBONACCI SERIES

AIM

To write a Fibonacci series program using Python language and to verify the sequence of generation.

SOFTWARE REQUIRED

Python 3.14

FORMULA USED

$$F_n = F_{n-1} + F_{n-2}$$

Where,

F_n - n^{th} term of Fibonacci series

F_{n-1}, F_{n-2} - previous terms

THEORY

The Fibonacci series is the sequence of numbers, where every number is the sum of the preceding two numbers, such that the first two terms are '0' and '1'. A Fibonacci series can thus be given as, 0, 1, 1, 2, 3, 5, 8, 13, 21, 34, . . . It can thus be observed that every term can be calculated by adding the two terms before it.

Given the first term, F_0 and second term, F_1 as '0' and '1' respectively, the third term is, $F_2 = F_0 + F_1 = 0 + 1 = 1$

Similarly,

$$F_3 = 1 + 1 = 2$$

$$F_4 = 2 + 1 = 3$$

$$F_5 = 2 + 3 = 5$$

$$F_6 = 3 + 5 = 8$$

$$F_7 = 5 + 8 = 13$$

and so, on

Therefore, to represent any $(n+1)^{\text{th}}$ term in this series, the general expression is, $F_n = F_{n-1} + F_{n-2}$.

PROBLEM

The number of leaves on a plant increases according to a recurring growth pattern based on the previous two stages. Write a Python program to generate the sequence for n stages.

APPLICATION PROBLEM

Write a Python program to verify the Source Transformation theorem by converting a voltage source with a series resistance into an equivalent current source with a parallel resistance, and vice versa, using the given circuit parameters.

ALGORITHM

STEP 1: Start

STEP 2: Input: Limit (the maximum value in the Fibonacci sequence)

STEP 3: Initialize variables:

- a. Fibonacci sequence: `sequence = [0, 1]`
- b. Current number: `current_num = 1`
- c. Previous number: `prev_num = 0`

STEP 4: While `current_num <= Limit`:

- a. Calculate the next Fibonacci number: `next_num = current_num + prev_num`
- b. If `next_num <= Limit`:
 - i. Append `next_num` to the Fibonacci sequence
 - ii. Update `prev_num` to `current_num`
 - iii. Update `current_num` to `next_num`
- c. Else, exit the loop

STEP 5: Construct a statement using the Fibonacci sequence:

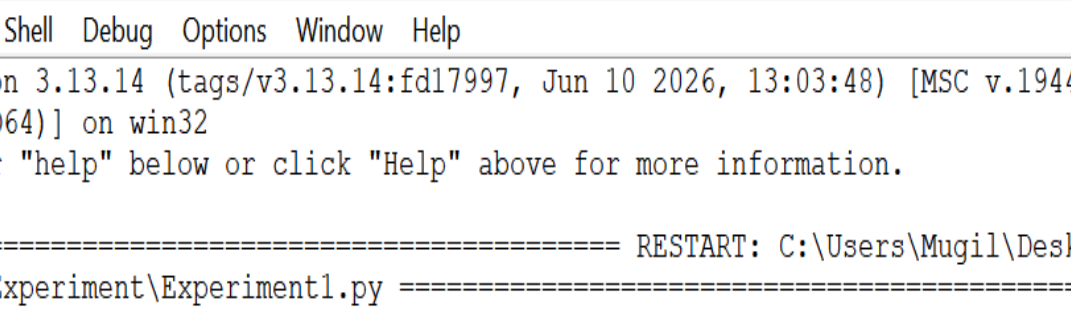
- a. If the Fibonacci sequence contains elements:
 - i. Construct a statement listing the Fibonacci sequence
- b. Else:
 - i. Output a message indicating that no Fibonacci numbers were found within the limit

STEP 6: Output the constructed statement

STEP 7: End

```
n = int(input("Enter the number of leaves: "))
firststage= 0
secondstage=1
print("Growth sequence:")
for i in range(n):
    print(firststage, end=" ")
    firststage = secondstage
    secondstage = firststage +secondstage
```

OUTCOMES OF PROGRAM CODING



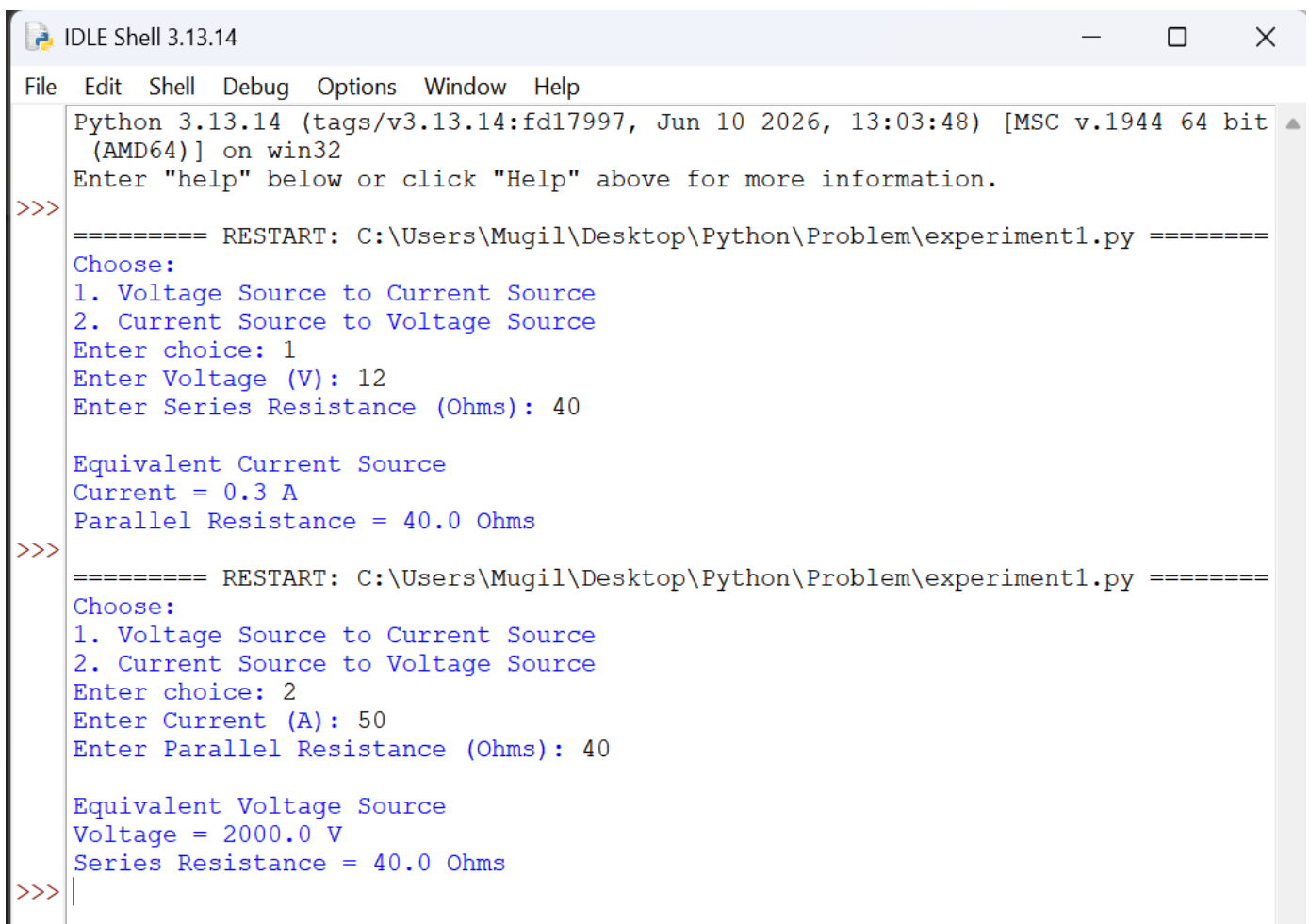
```
IDLE Shell 3.13.14
File Edit Shell Debug Options Window Help
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit
(AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:\Users\Mugil\Desktop\Pyt
hon\Experiment\Experiment1.py =====
Enter the number of leaves: 10
Growth sequence:
0 1 2 4 8 16 32 64 128 256
>>>
===== RESTART: C:\Users\Mugil\Desktop\Python\Experiment\Experiment1.py =====
Enter the number of leaves: 12
Growth sequence:
0 1 2 4 8 16 32 64 128 256 512 1024
>>> |
```

APPLICATION PROBLEM CODING

```

choice = int(input("Choose:\n1. Voltage Source to Current Source\n2. Current Source to
Voltage Source \nEnter choice: "))
if choice == 1:
    V = float(input("Enter Voltage (V): "))
    R = float(input("Enter Series Resistance (Ohms): "))
    I = V / R
    print("\nEquivalent Current Source")
    print("Current =", round(I, 2), "A")
    print("Parallel Resistance =", R, "Ohms")
elif choice == 2:
    I = float(input("Enter Current (A): "))
    R = float(input("Enter Parallel Resistance (Ohms): "))
    V = I * R
    print("\nEquivalent Voltage Source")
    print("Voltage =", round(V, 2), "V")
    print("Series Resistance =", R, "Ohms")
else:
    print("Invalid Choice")

```

OUTCOMES OF APPLICATION PROGRAM CODING


```

Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit
(AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:\Users\Mugil\Desktop\Python\Problem\experiment1.py =====
Choose:
1. Voltage Source to Current Source
2. Current Source to Voltage Source
Enter choice: 1
Enter Voltage (V): 12
Enter Series Resistance (Ohms): 40

Equivalent Current Source
Current = 0.3 A
Parallel Resistance = 40.0 Ohms
>>>
===== RESTART: C:\Users\Mugil\Desktop\Python\Problem\experiment1.py =====
Choose:
1. Voltage Source to Current Source
2. Current Source to Voltage Source
Enter choice: 2
Enter Current (A): 50
Enter Parallel Resistance (Ohms): 40

Equivalent Voltage Source
Voltage = 2000.0 V
Series Resistance = 40.0 Ohms
>>>

```

PROGRAM CODING

```
a = int(input("Enter first number: "))  
b = int(input("Enter second number: "))  
print("Sum =", a + b)  
print("Difference =", a - b)  
print("Product =", a * b)
```

OUTCOMES OF PROGRAM CODING

```
IDLE Shell 3.13.14  
File Edit Shell Debug Options Window Help  
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on  
win32  
Enter "help" below or click "Help" above for more information.  
>>>  
===== RESTART: C:\Users\Mugil\Desktop\Python\Problem1\1.py =====  
Enter first number: 5  
Enter second number: 10  
Sum = 15  
Difference = -5  
Product = 50
```

VIVA QUESTION**1. What is Python? List some popular applications of Python in the world of technology.**

Python is a high-level, general-purpose programming language known for its simplicity and readability. It supports multiple programming paradigms, including procedural, object-oriented, and functional programming.

Popular applications of Python:

- Web Development: Frameworks like Django and Flask.
- Data Science and Machine Learning: Libraries like NumPy, Pandas, and TensorFlow.
- Automation and Scripting: Writing automation scripts for system administration.
- Game Development: Libraries like Pygame.
- Artificial Intelligence (AI): Used in AI applications and research.
- Networking: Automation tools for network programming and cloud infrastructure management.

2. What are the benefits of using Python language as a tool in the present scenario?

- Easy to Learn and Use: Simple syntax that emphasizes readability.
- Wide Support for Libraries and Frameworks: Numerous libraries for various fields like data science, AI, and web development.
- Platform Independence: Can run on various platforms like Windows, macOS, and Linux.
- Community Support: Large and active community offering vast documentation and support.
- Integration and Versatility: Can be integrated with other languages and technologies.

3. Is Python a compiled language or an interpreted language?

Python is an interpreted language. This means that Python code is executed line by-line by the Python interpreter, unlike compiled languages that need the whole code to be translated into machine language before execution.

4. What does the '#' symbol do in Python?

The '#' symbol in Python is used to write comments. Comments are lines of text that the interpreter ignores during code execution, and they are primarily used for adding explanations or notes within the code.

5. What is the difference between a Mutable datatype and an Immutable data type?

Mutable Data Type: The objects of these data types can be changed after their creation.

Examples: list, dictionary, set.

Immutable Data Type: The objects of these data types cannot be changed once created.

Examples: tuple, string, integer.

RESULT:

Thus, the Python program successfully generated the Fibonacci sequence, executed correctly, and verified the sequence generation. Also, its application program was also executed and verified successfully.

INFERENCE:

The experiment demonstrated that Python effectively generates the Fibonacci sequence using iterative logic. It also confirmed the successful execution and verification of the application program, enhancing programming and problem-solving skills.

Assessment	Max. Marks	Marks Obtained
Conduction of Practical	15	
Report	5	
Viva	5	
Total Marks	25	
Signature of Lab In-Charge		